Sculpting Vulnerability Basins: Physical Sharpness-Aware Minimization for Embodied VLAs

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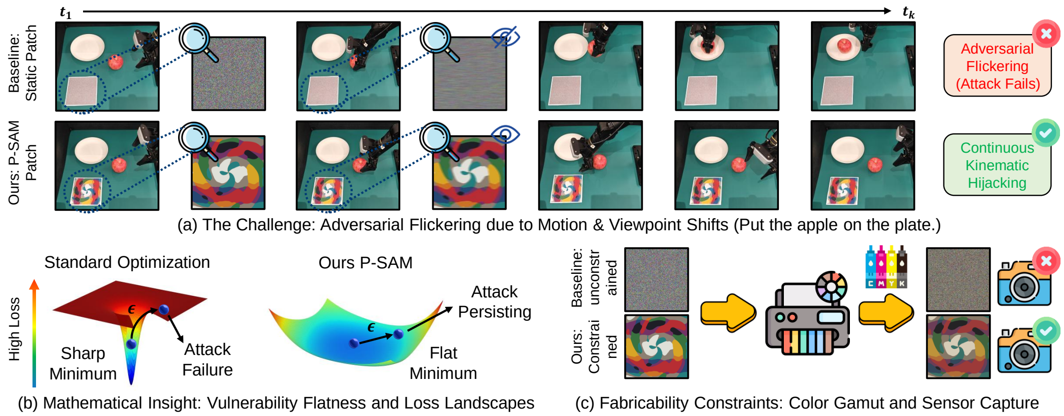
Abstract

Vision-Language-Action (VLA) models have emerged as powerful paradigms for embodied robotic control, yet their robustness against physical-world adversarial attacks remains underexplored. Current adversarial patch generation methods typically optimize over static, isolated 2D frames. However, deploying these patches against robotic systems exposes a critical flaw: standard optimizations tend to converge into fragile sharp minima. Consequently, these attacks suffer from severe adversarial flickering and fail completely when subjected to macroscopic kinematic shifts (e.g., continuous camera translation) and microscopic hardware degradations (e.g., printer gamut limits). To bridge this sim-to-real gap, we propose **P**hysical **S**harpness-**A**ware **M**inimization (**P-SAM**). Our framework mathematically decouples physical robustness into two tractable objectives. First, we model macroscopic trajectory dynamics using transformation expectations over offline datasets. Second, we utilize the white-box nature of the VLA to perform a minimax optimization, dynamically approximating the conditionally worst-case microscopic pixel noise via first-order Taylor expansions. By forcing gradient updates at these worst-case perturbed states, P-SAM actively sculpts broad vulnerability basins in the loss landscape. Combined with strict physical fabricability constraints, our method generates patches that maintain inescapable, devastating influence over the VLA's 7-DoF kinematics. Evaluations across diverse manipulation tasks demonstrate that P-SAM significantly outperforms baseline static patches, underscoring an urgent need for robust defensive paradigms in embodied AI. Our project can be found at https://psam-26.github.io.

1 Introduction

The integration of high-capacity vision encoders with large language models [Deitke et al., 2025, Beyer et al., 2024, Chen et al., 2023, Qwen et al., 2025, Karamcheti et al., 2024, Alayrac et al., 2022, Wang et al., 2024, Yao et al., 2024] has catalyzed a paradigm shift in embodied artificial intelligence. Vision-Language-Action (VLA) foundation models [Wang et al., 2025b, Kim et al., 2024, Black et al., 2026, Ding et al., 2025, Shi et al., 2025, Team et al., 2025] can directly map raw visual observations and natural language instructions to continuous, low-level robotic kinematics (e.g., 7-Degree-of-Freedom control). By fusing semantic reasoning with physical execution in an end-to-end manner, VLAs enable robots to perform complex, zero-shot manipulation tasks in unstructured, open-world environments. However, as these models transition from simulated benchmarks to safety-critical physical deployments [Liu et al., 2025, NVIDIA et al., 2025, Intelligence et al., 2025, Smith and Coles, 1973, Kroemer et al., 2020, Ingrand and Ghallab, 2017], their direct reliance on unmediated visual inputs introduces a profound attack surface. Understanding and mitigating their vulnerabilities against malicious physical-world adversarial interference has consequently emerged as a paramount security imperative.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.



(a) The Challenge: Adversarial Flickering due to Motion & Viewpoint Shifts (Put the apple on the plate.)

(b) Mathematical Insight: Vulnerability Flatness and Loss Landscapes

(c) Fabricability Constraints: Color Gamut and Sensor Capture

Figure 1: Overcoming adversarial flickering in embodied VLAs. (a) Standard patches optimized on static frames fail in closed-loop systems, as macroscopic kinematic shifts and microscopic hardware noise instantly neutralize the attack. (b) While existing methods converge to fragile sharp minima (left), our P-SAM sculpts robust flat minima (right), maintaining attack efficacy despite unmodeled physical perturbations ϵ. (c) TV and NPS constraints ensure physical fabricability, preventing color clipping and guaranteeing real-world deployment.

Historically, physical adversarial attacks (e.g., printable patches placed in the environment) have proven highly effective at deceiving static computer vision models [Eykholt et al., 2018a,b, Wei et al., 2024, DelaCruz et al., 2026, Nguyen et al., 2024, Wang et al., 2023]. Yet, deploying these conventional patches against an embodied, closed-loop robotic system exposes a fundamental methodological flaw. Current adversarial patch generation methods [Wang et al., 2025a, Xu et al., 2020a, Huang et al., 2026] typically optimize perturbations over isolated, static 2D frames. However, a robotic manipulator operates as a continuous dynamical system. As the robot moves, the patch undergoes macroscopic kinematic shifts (e.g., continuous changes in camera translation, scale, and viewing angle) and is subjected to microscopic hardware degradations (e.g., CMOS sensor grain, motion blur, and printer color gamut clipping). Because standard empirical risk minimization naturally converges into highly localized, sharp minima in the highly non-convex adversarial loss landscape, even a microscopic deviation instantly ejects the patch from this fragile optimal state. We characterize this phenomenon as adversarial flickering (see Figure 1 (a)), a state where the malicious patch succeeds in one frame but shatters the moment the robot arm moves, allowing the VLA's closed-loop feedback mechanism to safely auto-correct its trajectory and override the attack.

Bridging this adversarial sim-to-real gap poses a formidable technical bottleneck. Naive solutions might attempt to simulate the continuous temporal dynamics of the robot using heavy 3D differentiable physics simulators or by unrolling full kinematic trajectories during optimization. However, such approaches introduce prohibitive computational overhead, complex sim-to-real dynamics modeling, and severe memory bottlenecks, rendering them highly impractical for scaling across diverse robotic embodiments and environments.

To overcome these challenges without the computational explosion of 3D digital twins, we propose **P**hysical **S**harpness-**A**ware **M**inimization (**P-SAM**). We hypothesize that for a physical attack to reliably hijack a continuous control loop, the optimization must actively sculpt a broad, robust vulnerability basin (i.e., a flat minimum, as illustrated in Figure 1 (b)) rather than a sharp valley. Our computationally lightweight framework mathematically decouples physical robustness into two tractable objectives entirely within the 2D domain. First, to account for macroscopic kinematic shifts, we utilize Expectation over Transformation (EoT) on single frames sampled from offline robotic datasets, cheaply simulating the continuous camera motion of the robot arm traversing 3D space. Second, to combat microscopic hardware noise, P-SAM formulates a minimax objective utilizing the white-box computational graph of the VLA. We dynamically approximate the conditionally worst-case pixel perturbation for any given macroscopic perspective via a first-order Taylor expansion, requiring only a single backward pass. By forcing the gradient descent step to evaluate at these worst-case perturbed states, our optimization acts as a mathematical repellent, dragging the patch parameters away from fragile cliffs and explicitly flattening the adversarial loss landscape.

To guarantee that this mathematically optimal flat minimum translates seamlessly to the physical world, we strictly constrain the optimization using Total Variation (TV) and Non-Printability Score

(NPS) regularizers. These constraints ensure the resulting patches feature organic, smooth color transitions that survive motion blur and are perfectly reproducible by standard commercial printers (see Figure 1 (c)). The final patches maintain an inescapable, devastating influence over the VLA's closed-loop kinematics, bypassing the need for digital access to the target robot.

In summary, our core contributions are as follows:

- We expose a critical failure mode of current static adversarial patches against VLAs, mathematically attributing the physical sim-to-real gap to the convergence of standard optimizers into brittle sharp minima.
- We propose a novel, computationally lightweight framework that elegantly decouples macroscopic robotic kinematics (via offline 2D EoT) from microscopic hardware noise. By forcing gradient updates at conditionally worst-case perturbations, P-SAM actively sculpts broad vulnerability basins without requiring heavy 3D differentiable rendering.
- By integrating P-SAM with strict physical fabricability constraints, we generate adversarial patches that reliably and consistently hijack continuous 7-DoF robotic manipulation tasks in the real world. Extensive evaluations demonstrate that our method drastically outperforms existing baseline static patches, underscoring an urgent need for robust defensive architectures in next-generation VLA models.

2 Method

Current adversarial patch generation methods for Vision-Language-Action (VLA) models predominantly optimize perturbations over static, isolated 2D images. This static assumption fundamentally ignores the dynamic realities of embodied robotic systems. During real-world execution, an adversarial patch is continuously subjected to macroscopic kinematic shifts (e.g., motion distortion, scaling, and perspective changes as the camera moves) and microscopic hardware degradations (e.g., CMOS sensor noise, ambient lighting jitter, and printer color gamut clipping). By ignoring these physical factors, conventional static 2D optimizations converge into brittle solutions that suffer from adversarial flickering, a phenomenon where a patch succeeds in a pristine digital frame but fails the moment the robotic arm moves.

To bridge this critical sim-to-real gap, we propose a novel framework that systematically embeds physical-world dynamics into the 2D optimization pipeline (as shown in Figure 2). Our method mathematically decouples physical robustness into three tractable objectives: simulating macroscopic motion dynamics (Section 2.1), sculpting flat minima to absorb microscopic hardware noise (Section 2.2), and bounding the optimization with strict physical fabricability constraints to guarantee zero-shot real-world transferability (Section 2.3).

2.1 Macroscopic Kinematic Simulation via Expectation over Transformation

Conventional adversarial attacks generate patches by overfitting to pristine, static viewpoints. The moment the robotic arm initiates a trajectory, the resulting perspective distortion (changes in scale, viewing angle, and spatial warping) instantly breaks the adversarial alignment. To guarantee robustness against these macroscopic kinematic shifts without leaving the efficient 2D domain, we employ Expectation over Transformation (EoT) across offline workspace datasets (see Figure 2 (a)).

Formally, the physically augmented observation fed into the VLA is defined as:

$$o(x, \delta, \tau) = (1 - M_\tau) \odot x + M_\tau \odot \mathcal{T}(\delta, \tau), \tag{1}$$

where $x \in \mathbb{R}^{H \times W \times 3}$ is a clean, benign camera frame sampled from an offline dataset (e.g., LIBERO), and δ represents the learnable parameters of the 2D adversarial patch. To simulate the continuous perspective distortions and motion artifacts caused by the robot's movement, we sample a physical transformation τ from a predefined physical distribution Γ (encompassing random affine scaling, perspective warping, rotation, and brightness shifts). The function $\mathcal{T}(\delta, \tau)$ applies this macroscopic transformation to the digital patch δ, while the operator $\odot$ denotes the Hadamard (element-wise) product.

Crucially, the dynamically transformed binary mask $M_\tau \in \{0, 1\}^{H \times W}$ is not merely a spatial bounding box, but the mathematical guarantor of physical realizability. To distinguish a legitimate

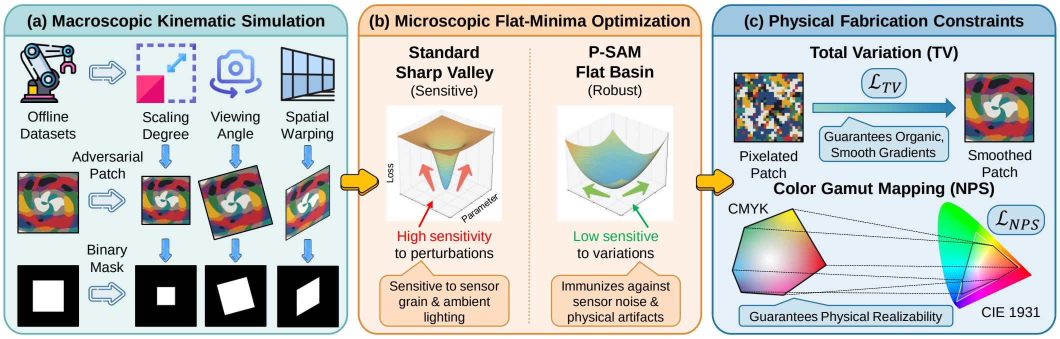


Figure 2: Overview of the P-SAM framework. Our method mitigates the adversarial sim-to-real gap via three decoupled 2D objectives. (a) Macroscopic Kinematic Simulation: Expectation over Transformation (EoT) with dynamic opaque masking models continuous 3D perspective shifts. (b) Microscopic Flat-Minima Optimization: Evaluating gradients at worst-case perturbed states sculpts a flat vulnerability basin to mitigate hardware-induced adversarial flickering. (c) Physical Fabricability Constraints: $\mathcal{L}_{TV}$ and $\mathcal{L}_{NPS}$ regularizers ensure the patch survives motion blur and CMYK gamut clipping during physical deployment.

localized physical attack from an impossible global digital perturbation, M_τ enforces two strict constraints: opaque occlusion and dynamic geometric synchronization. To prevent physically impossible semi-transparent overlays, $(1 - M_\tau) \odot x$ acts as a spatial carving mechanism that precisely deletes background pixels before patch insertion. Simultaneously, applying the identical macroscopic transformation τ to the base mask ensures that M_τ tracks the patch's perspective distortions. This prevents out-of-bound digital artifacts and strictly confines adversarial gradients to the valid physical dimensions of the deployable sticker. A comprehensive mathematical and geometric analysis of these constraints is provided in Appendix C.3.

Given this strictly bounded, physically augmented observation, we formulate the base targeted manipulation loss:

$$\mathcal{L}_{base}(\delta, x, \tau) = \mathcal{L}_{CE}\Big(\mathcal{F}_{VLA}\big(o(x, \delta, \tau), I\big),\ A_{target}\Big), \tag{2}$$

where $\mathcal{F}_{VLA}(\cdot)$ acts as a frozen white-box VLA model, whose internal weights are fixed, but full access to its computational graph is maintained to backpropagate gradients. $\mathcal{L}_{CE}$ denotes the cross-entropy loss, which quantifies the discrepancy between the VLA's predicted action distribution (conditioned on the natural language instruction I) and the attacker's malicious 7-DoF target A_{target}.

2.2 Conquering Microscopic Noise via Physical Sharpness-Aware Minimization

While EoT addresses macroscopic viewpoint changes, standard empirical risk minimization over static datasets inevitably drives the patch parameters into a sharp minimum. In the physical world, cameras do not capture perfect pixel matrices. Unmodeled microscopic variations (e.g., CMOS camera sensor grain, ambient lighting fluctuations, and slight printer ink bleeding) act as random noise vectors that instantly eject the patch from this fragile minimum, neutralizing the attack.

To actively sculpt a globally robust vulnerability basin (i.e., a flat minimum) that immunizes the patch against these unpredictable sensor and environmental noises (as shown in Figure 2 (b)), we introduce Physical Sharpness-Aware Minimization (P-SAM). We enforce a constraint that the patch must remain aggressively malicious even when subjected to the absolute worst-case microscopic pixel degradation. This is formulated as a minimax objective:

$$\min_{\delta} \mathbb{E}_{x,\tau}\left[\max_{\|\epsilon\|_2 \le \rho} \mathcal{L}_{base}(\delta + \epsilon, x, \tau)\right], \tag{3}$$

where ϵ is a microscopic, learnable perturbation applied strictly to the patch pixels, acting as a mathematical surrogate for unmodeled physical hardware noise. The hyperparameter ρ defines the ℓ_2-norm radius of the perturbation ball, effectively controlling the desired width of the vulnerability basin.

Solving the exact inner maximization iteratively is computationally intractable for deep neural networks. However, utilizing the white-box gradients of the VLA, we can efficiently approximate the conditionally worst-case noise ϵ^* via a first-order Taylor expansion (see Appendix C.2 for the detailed mathematical derivation and physical justification), requiring only a single additional backward pass per iteration:

$$\epsilon^*(x,\tau) = \rho \frac{\nabla_\delta \mathcal{L}_{base}(\delta, x, \tau)}{\|\nabla_\delta \mathcal{L}_{base}(\delta, x, \tau)\|_2}, \tag{4}$$

where $\nabla_\delta \mathcal{L}_{base}$ denotes the standard gradient of the base loss with respect to the patch parameters. Crucially, this worst-case pixel noise $\epsilon^*(x,\tau)$ is strictly conditioned on the specific background image x and the sampled macroscopic perspective τ.

Instead of updating the patch using the gradient at its current state δ_k, P-SAM computes the gradient at this worst-case perturbed state. The patch parameters are thus updated as:

$$\delta_{k+1} = \delta_k - \alpha \cdot \mathbb{E}_{(x,\tau)\sim\mathcal{B}} \left[\nabla_\delta \mathcal{L}_{base}\big(\delta_k + \epsilon^*(x,\tau), x, \tau\big) \right], \tag{5}$$

where $\mathcal{B}$ is the current mini-batch of sampled image-transformation pairs and α is the optimization learning rate.

Geometric Interpretation of Global Flatness: As the local loss landscape varies continuously with respect to the macroscopic viewpoint τ, the corresponding worst-case perturbation ϵ^* exhibits significant variance. By independently computing the conditional worst-case perturbation for each (x,τ) pair within a mini-batch and averaging their perturbed gradients, the optimization trajectory of the patch δ simultaneously minimizes the loss across an ensemble of viewpoint-specific worst-case degradations. This dynamic effectively regularizes against sharp local minima, encouraging convergence toward a flatter, more resilient vulnerability basin (see Figure 3).

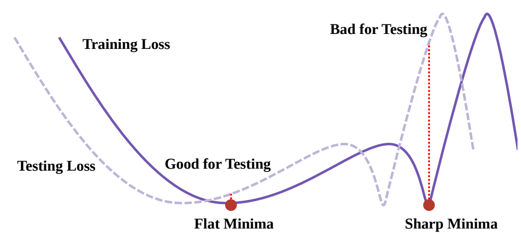


Figure 3: Comparison of flat and sharp minima under physical domain shifts. The domain shift is conceptually simulated by curve translation.

2.3 Physical Fabricability Constraints

Finally, a mathematically optimal digital patch is fundamentally useless if it cannot be accurately reproduced in the physical world. Static 2D optimization methods inherently assume perfect digital-to-physical pixel fidelity, frequently generating highly saturated, high-frequency noise patterns. Such patterns not only exceed the color gamut of physical printers (resulting in severe color clipping) but are also instantly destroyed by the motion blur inherent to a moving robotic camera. To bridge this final deployment gap, we enforce two strict physical regularizers (as shown in Figure 2 (c)).

First, to penalize high-frequency color transitions and ensure the adversarial signal survives motion blur and camera down-sampling, we apply Total Variation (TV) regularization:

$$\mathcal{L}_{TV}(\delta) = \sum_{i,j} \sqrt{(\delta_{i+1,j} - \delta_{i,j})^2 + (\delta_{i,j+1} - \delta_{i,j})^2}, \tag{6}$$

which guarantees organic, smooth gradients across the spatial coordinates (i,j) of the patch.

Second, to ensure the digital pixels can be faithfully reproduced by physical ink without distortion, we apply the Non-Printability Score (NPS):

$$\mathcal{L}_{NPS}(\delta) = \sum_{p\in\delta} \min_{c\in C} \|p - c\|_2, \tag{7}$$

which strictly binds every digital pixel vector p to the closest matching color c within a predefined, physically reproducible CMYK printer color gamut C. This prevents unpredictable out-of-gamut color clipping during fabrication.

Final Unified Objective. Integrating the macroscopic kinematic simulation (EoT), the microscopic flat-minima optimization (P-SAM), and the physical fabricability constraints, the complete, end-to-end objective function optimized by the attacker is:

$$\mathcal{L}_{final}(\delta) = \mathbb{E}_{(x,\tau)\sim\mathcal{B}} \Big[\mathcal{L}_{base}\big(\delta + \epsilon^*(x,\tau), x, \tau\big) \Big] + \lambda_1 \mathcal{L}_{TV}(\delta) + \lambda_2 \mathcal{L}_{NPS}(\delta), \tag{8}$$

Algorithm 1 Physical Sharpness-Aware Minimization (P-SAM)

Input: Offline dataset $\mathcal{D}$, Frozen VLA $\mathcal{F}_{VLA}$ with language instruction I, Target Action A_{target}
Parameters: Learning rate α, perturbation bound ρ, constraint weights λ_1, λ_2
Initialize: Patch parameters δ_0

```
for step k = 0, 1, 2, ... do
    Sample a mini-batch of clean frames x ~ D and physical transforms τ ~ Γ
    Construct augmented observations o = (1 − M_τ) ⊙ x + M_τ ⊙ T(δ_k, τ)
    // Step 1: Inner Maximization (Approximate worst-case noise)
    Compute base loss L_base = L_CE(F_VLA(o, I), A_target)
    Calculate worst-case perturbation: ε* = ρ ∇_δ L_base / ||∇_δ L_base||_2
    // Step 2: Outer Minimization (Evaluate at perturbed state)
    Apply worst-case noise to the patch: δ̂ = δ_k + ε*
    Construct noisy observation o_noisy = (1 − M_τ) ⊙ x + M_τ ⊙ T(δ̂, τ)
    Compute perturbed loss L̂_base = L_CE(F_VLA(o_noisy, I), A_target)
    Compute gradients of constraints: ∇_δ L_TV(δ_k) and ∇_δ L_NPS(δ_k)
    // Step 3: Apply Flat-Minima Update
    Compute final gradient: g = ∇_δ L̂_base + λ_1 ∇_δ L_TV(δ_k) + λ_2 ∇_δ L_NPS(δ_k)
    Update patch: δ_{k+1} = δ_k − α · g
    Project δ_{k+1} to valid RGB color space [0, 1]
end for
Return: Optimized, physically robust adversarial patch δ*
```

where λ_1 and λ_2 are empirically tuned weighting coefficients that balance the targeted kinematic hijacking success rate against the stealth and physical printability of the adversarial patch. The full optimization procedure is summarized in Algorithm 1. By optimizing Equation 8 entirely offline, the attacker bypasses the need for computationally prohibitive 3D simulations or real-time queries to the target robot. The resulting P-SAM patch acts as a condensed vulnerability basin, inherently robust to both macroscopic kinematic shifts and microscopic physical degradations.

3 Experiments

3.1 Experimental Setup

Victim Models and Datasets. We evaluate our framework against OpenVLA-7B [Kim et al., 2024] and π_0 Black et al. [2026], prominent examples of open-weights Vision-Language-Action foundation models. LIBERO [Liu et al., 2023] is a simulation dataset designed to evaluate models across four distinct task types (Spatial, Object, Goal, Long). Notably, LIBERO-Spatial emphasizes spatial reasoning and geometric understanding, requiring models to comprehend relative positions and scene layouts, while LIBERO-Long combines diverse objects, layouts, and long-horizon tasks, making it challenging for complex, multi-step planning. Therefore, we utilize both the LIBERO-Long and LIBERO-Spatial suites to comprehensively assess these capabilities.

Baselines and Objectives. To systematically decompose the physical sim-to-real gap and validate the necessity of each component in our framework, we establish the following contrastive baselines and ablation configurations. All optimized patches utilize a targeted manipulation objective to maximize the continuous kinematic deviation from the benign trajectory: **(1) Clean (Initial Model):** Normal robotic execution without any adversarial interference, establishing the foundational capabilities of the VLA. **(2) Random Noise:** A patch instantiated with pixel values uniformly sampled from a Gaussian distribution, evaluating the VLA's natural robustness against unstructured visual clutter. **(3) w/o Viewpoint Deformation:** A static adversarial patch optimized via standard Projected Gradient Descent (PGD) on isolated 2D frames, strictly ignoring continuous macroscopic kinematic shifts. **(4) w/o Flat Minima:** A patch that incorporates macroscopic Expectation over Transformation (EoT) but omits our microscopic minimax optimization, thus naturally converging to a fragile sharp minimum. **(5) w/o $\mathcal{L}_{TV}$:** The P-SAM framework excluding the Total Variation regularizer, permitting high-frequency spatial pixel transitions. **(6) w/o $\mathcal{L}_{NPS}$:** The P-SAM framework excluding the Non-Printability Score constraint, freely generating out-of-gamut digital colors. **(7) Ours (Full P-SAM):** Our complete, unified framework encompassing macroscopic EoT, microscopic worst-case minimax optimization, and strict physical fabricability constraints.

Table 1: Comparison of P-SAM and baselines. In Real-World scenarios, we employ (TFR ↑) as the evaluation metric. OBJ, PLC, and MNP denote object grasping, placement, and manipulation.

Method Configuration	Model	LIBERO-Long			LIBERO-Spatial			Real-World		
		TFR (%)	NAD (%)	ASR (%)	TFR (%)	NAD (%)	ASR (%)	OBJ (%)	PLC (%)	MNP (%)
Clean (Initial Model)	Openvla	46.3	-	-	15.3	-	-	12	14	20
Random Noise		48.4	-	-	16.6	-	-	14	17	21
w/o Viewpoint Deformation		60.6	30.8	19.5	38.2	16.8	8.9	37	43	48
w/o Flat Minima		86.8	42.7	29.0	71.6	26.1	18.3	40	49	45
w/o $\mathcal{L}_{TV}$ (No Smoothness)		72.4	34.9	25.4	49.2	14.5	10.7	35	44	47
w/o $\mathcal{L}_{NPS}$ (No Gamut Sync)		98.2	61.5	44.6	88.4	45.3	36.9	26	28	29
Ours (Full P-SAM)		**100**	**63.6**	**47.1**	**95.6**	**50.7**	**38.5**	**62**	**71**	**76**
Clean (Initial Model)	π_0	6.2	-	-	1.0	-	-	1	4	7
Random Noise		8.2	-	-	1.8	-	-	2	4	9
w/o Viewpoint Deformation		31.4	12.5	6.2	18.8	8.3	4.5	16	20	26
w/o Flat Minima		37.8	16.4	9.1	35.2	15.9	8.7	29	28	30
w/o $\mathcal{L}_{TV}$ (No Smoothness)		36.0	16.1	9.0	36.2	16.2	8.8	27	29	34
w/o $\mathcal{L}_{NPS}$ (No Gamut Sync)		59.0	25.7	16.8	54.4	23.6	16.4	11	17	22
Ours (Full P-SAM)		**72.6**	**28.9**	**20.5**	**71.6**	**29.3**	**20.2**	**43**	**45**	**58**

Evaluation Metrics. Unlike static image classification, evaluating robotic hijacking requires continuous temporal metrics: **(1) Task Failure Rate (TFR ↑):** The percentage of task rollouts that fail to complete the designated instruction. **(2) Normalized Action Deviation (NAD ↑):** The continuous ℓ_2-norm distance between the adversarially predicted 7-DoF actions and the unperturbed ground-truth trajectory, normalized by the empirical operational range of each DoF. This strictly quantifies the true severity of the ongoing spatial and rotational hijacking. **(3) Attack Sustaining Rate (ASR ↑):** The percentage of sequential frames within a rollout where the adversarial patch successfully induces an NAD greater than a critical safety threshold ($\beta = 0.1$). A low ASR indicates the attack repeatedly loses its grip, allowing the VLA to auto-correct and return to its benign path. See Appendix A and Appendix E for more details.

3.2 Main Result

Table 1 summarizes quantitative evaluations across continuous noisy simulations and real-world physical deployments. While **Clean** models and **Random Noise** confirm the inherent resilience of VLAs against unstructured visual anomalies, conventional adversarial patches expose a profound sim-to-real gap. For instance, the static **w/o Viewpoint Deformation** baseline suffers a catastrophic real-world collapse (e.g., ASR plunges to 19.5% on OpenVLA and 6.2% on π_0), empirically capturing adversarial flickering: static attacks instantly misalign during continuous camera translation. Although the **w/o Flat Minima** patch survives macroscopic shifts, its sharp optimum remains brittle to unmodeled hardware noise, allowing the VLA to auto-correct and stalling real-world TFRs at 40%–49% for OpenVLA. Furthermore, digital robustness is useless without fabrication constraints. The **w/o** $\mathcal{L}_{NPS}$ baseline exhibits a textbook sim-to-real collapse (98.2% digital to 26%–29% physical TFR) due to irreversible CMYK color clipping, while the **w/o** $\mathcal{L}_{TV}$ patch generates high-frequency static that is easily destroyed by natural motion blur acting as a low-pass filter.

In stark contrast, **Ours (Full P-SAM)** acts as a mathematical shock absorber. By actively sculpting a physically bounded flat vulnerability basin, P-SAM sustains a 100% digital TFR on OpenVLA-Long and maximizes real-world TFRs up to 76% in complex Manipulation (MNP) tasks. Even against the resilient π_0 architecture, P-SAM successfully scales the physical failure rate to 58%. Crucially, P-SAM nearly doubles the ASR compared to baseline methods (e.g., reaching 47.1% on OpenVLA-Long), ensuring that the continuous adversarial hijacking remains inescapable throughout the physical execution.

3.3 More Analysis

Spatial Sensitivity Tests. A critical dimension of real-world threat modeling is whether an adversarial patch requires strict physical proximity to the target object to succeed. Taking OpenVLA as the target model, we systematically varied the patch placement across nine distinct angular directions (Figure 4 (a)) and incrementally increased its distance offset up to 30 centimeters (Figure 4 (b)). Strikingly, P-SAM maintains a consistently high TFR across all angular zones, experiencing negligible degradation even when pushed to the peripheral boundaries of the workspace. Because modern VLA vision encoders rely heavily on global self-attention mechanisms to construct holistic scene representations, the optimized patch effectively acts as an "attention sink" that corrupts

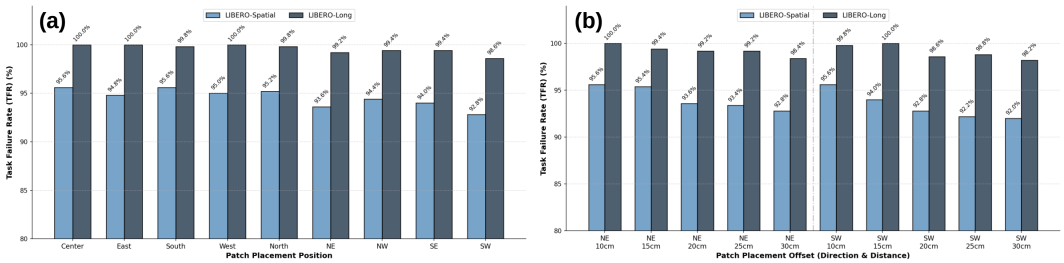


Figure 4: TFR across different spatial patch placements. **(a)** TFR across diverse angular placements. **(b)** TFR under increasing distance offsets.

the global visual feature space from any visible location. This spatial and distance invariance underscores the stealth of the P-SAM attack, explicitly demonstrating that adversaries can deploy the patch in highly inconspicuous peripheral regions without compromising continuous attack efficacy.

Hyperparameter Evaluations. The perturbation radius ρ in our minimax objective dictates the width of the sculpted vulnerability basin. As shown in Figure 5 (a), increasing ρ from 0.01 to 0.05 significantly raises the real-world TFR (e.g., scaling up to 76% in OpenVLA manipulation tasks and 71% in OpenVLA placement tasks), demonstrating enhanced robustness against noise. However, overly aggressive bounds ($\rho > 0.1$) excessively restrict the optimization space, leading to underfitting and a subsequent decline in overall targeted efficacy. We empirically select $\rho = 0.05$ as the optimal trade-off between physical resilience and adversarial severity. We also investigate the relationship between the adversarial patch size and hijacking efficacy in Figure 5 (b). We observe a direct correlation: as the patch size increases from 2% to 10% of the observation frame, the TFR rises across both models. While larger patches provide a broader canvas for the optimizer to embed robust spatial features, we constrain our default patch size to 5% to strike a balance between environmental stealth and critical system disruption. More supplementary experiments can be found in Appendix D.

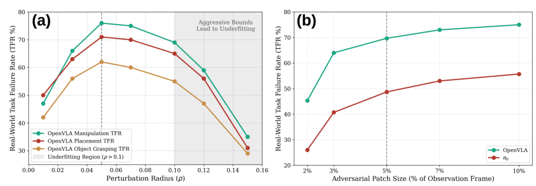


Figure 5: The TFR results of ρ and patch size. The result of the patch size is the average value of OBJ, PLC and MNP.

Transferability Assessments. In realistic security threats, attackers rarely possess perfect white-box access to the deployed VLA's internal parameters. Optimization theory suggests that converging to flat minima mitigates overfitting to the surrogate model's specific decision boundaries, thereby inherently enhancing adversarial transferability. To rigorously evaluate this black-box threat potential, we conducted extensive zero-shot evaluations across both distinct architectures and diverse environments. Specifically, we optimized the adversarial patch on a single model and deployed it against other models, including OpenVLA [Kim et al., 2024], π_0 [Black et al., 2026], $\pi_{0.5}$ [Intelligence et al., 2025], and FAST [Pertsch et al., 2025] (see Figure 6 (a)). Furthermore, we investigated cross-scenario transferability by generating the patch within a specific setting and assessing its attack efficacy across distinct LIBERO [Liu et al., 2023] task suites (Spatial, Object, Goal, and Long) (see Figure 6 (b)). We can find that our P-SAM maintained zero-shot physical transferability. By sculpting a wide vulnerability basin, P-SAM captures generalized semantic flaws in the underlying vision encoders, posing a realistic physical threat even under strict black-box and cross-domain assumptions.

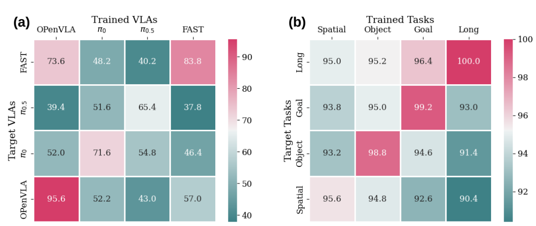


Figure 6: The TFR results of transferability across different VLAs and tasks. (a) Testing on LIBERO-Spatial. (b) Testing on OpenVLA.

Qualitative Demonstrations. To intuitively illustrate the behavioral shifts induced by different patches, Figure 7 and Figure 8 provides visualizations of continuous execution. **(1) Simulation Instance:** In a "Pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate" task, the conventional adversarial attack baseline causes an initial adversarial deviation. However, as the camera moves, perspective distortion and simulated noise trigger adversarial flick-

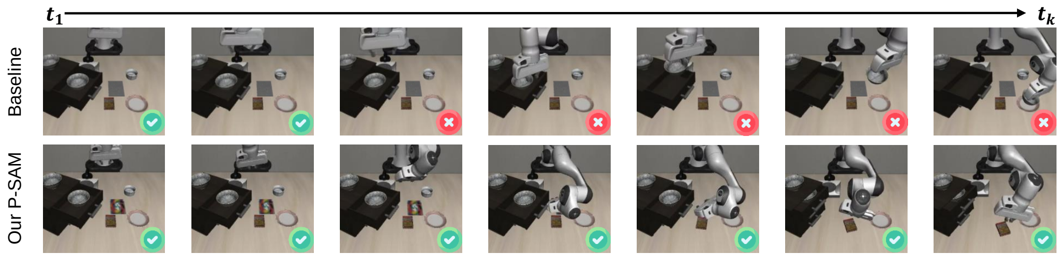


Figure 7: Qualitative results of the simulated scenario. The deployed model is OpenVLA.

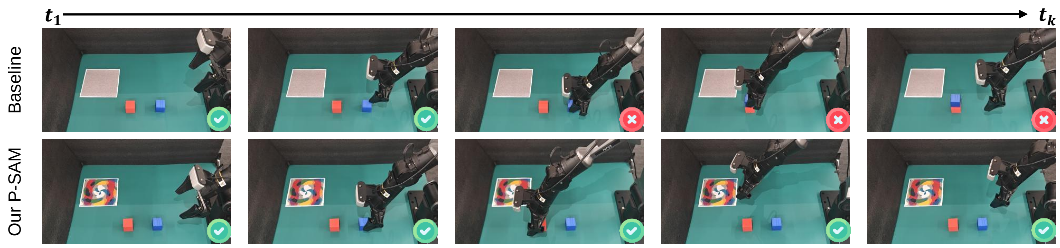


Figure 8: Qualitative results of the real-world scenario. The deployed model is OpenVLA.

ering; the VLA regains perception, auto-corrects the trajectory, and successfully completes the task. Conversely, P-SAM maintains a relentless adversarial grip, forcing the arm into a persistent erratic trajectory. **(2) Real-World Instance:** Deployed on the PiPER arm for a "Place the blue square on top of the red one" task, the conventional adversarial attack baseline patch fails within the first few seconds due to ambient lighting shifts and motion blur. However, the P-SAM patch effectively overrides the intended semantic instruction and subverts the 7-DoF closed-loop kinematics, thereby exposing critical physical security vulnerabilities. Appendix G presents more results.

4 Related Work

Adversarial attacks serve as indispensable tools for rigorously evaluating model vulnerabilities in the dynamic physical environments where robotic systems operate. While conventional gradient-driven pixel-level perturbations [Wang et al., 2019, Su et al., 2019, Madry et al., 2019, Liu et al., 2019, Goodfellow et al., 2015, Du et al., 2021] achieve high efficacy in static digital simulations, they fundamentally lack physical viability. Consequently, patch-based adversarial techniques [Wu et al., 2025, Xu et al., 2020b, Chen et al., 2023, Brown et al., 2018, Athalye et al., 2018, Billard and Kragic, 2019] have emerged as a more practical alternative by generating localized and physically fabricable perturbations. These patches remain robust against environmental shifts like changing illumination and camera angles. Given that Vision-Language-Action (VLA) models are designed for real-world execution, assessing their operational security through physical patches is highly critical.

5 Conclusion

In this work, we identified adversarial flickering as a primary cause of physical attack failures against Vision-Language-Action (VLA) models, demonstrating that static patches inherently converge into fragile, sharp minima. To address this, we proposed P-SAM, a framework that actively sculpts flat vulnerability basins by coupling macroscopic 2D perspective augmentations with a conditionally worst-case microscopic minimax optimization. Enforced by strict fabricability constraints, P-SAM generates deployable patches capable of reliably hijacking continuous robotic manipulation tasks in the physical world. By exposing critical vulnerabilities in VLA models that caution against impetuous real-world deployments, our framework serves as a pioneering contribution toward the reliability of AI-enhanced robotic systems.

Limitations: First, our offline patch generation process strictly requires white-box gradient access to the target VLA or an aligned surrogate to compute the worst-case pixel perturbations. Second, the physical attack inherently relies on continuous camera line-of-sight, meaning severe topological occlusions will temporarily suspend the adversarial influence.

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A Extended Experimental and Implementation Details

To ensure the full reproducibility of our Physical Sharpness-Aware Minimization (P-SAM) framework and to provide deeper insights into the physical deployment constraints, we elaborate on the intricate experimental setups, hardware specifications, and optimization hyperparameters omitted from the main text due to space limitations.

A.1 Evaluation Details

(1) Simulation Setting: Each suite consists of 10 tasks, with each task executed for 50 trials, resulting in a total of 500 rollouts, following [Kim et al., 2024]. For reproducibility, we carefully selected patch paste locations for each suite, ensuring they did not obscure objects or robots in the test environment. To emulate physical degradation digitally, we injected Gaussian sensor noise ($\sigma = 0.05$) and dynamic motion blur to the camera observations at each timestep. **(2) Real-World Setting:** We deployed VLAs on a physical 7-DoF PiPER robotic arm equipped with a Robotiq Hand-E Gripper and a static RGB shoulder camera. The evaluation encompassed 100 trials for each of the three distinct tasks: object grasping, placement, and manipulation.

A.2 Compute Platform and Hardware Infrastructure

Offline Optimization Platform. All digital adversarial optimizations, including the macroscopic Expectation over Transformation (EoT) sampling and the microscopic minimax gradient computations (P-SAM), were conducted on a dedicated computing cluster equipped with NVIDIA RTX 5880 GPUs (48GB VRAM per GPU) to accommodate the intensive memory requirements.

Software Stack. The optimization pipeline was implemented on Ubuntu 22.04 LTS using Python 3.10 and PyTorch 2.3.0 with CUDA Toolkit 12.1. During optimization, frozen VLA models such as OpenVLA-7B were loaded via the Hugging Face transformers and accelerate libraries, while the diffusion-based π_0 was deployed using its respective custom environment. All model forward passes and gradient computations were executed in bfloat16 precision. This precision choice optimally balances VRAM utilization while strictly preserving gradient fidelity (preventing underflow) during the deep backward pass to the image space. Furthermore, FlashAttention-2 and gradient checkpointing were enabled to significantly accelerate the attention mechanisms within the visual encoders (e.g., SigLIP) and LLM backbones, while strictly containing the activation memory footprint. Finally, spatial physical augmentations were hardware-accelerated using the kornia library, keeping the entire EoT (Expectation over Transformation) pipeline fully differentiable and strictly on the GPU without CPU-transfer bottlenecks.

A.3 Real-World Robotic Platform Configuration

Robotic Manipulator. We utilized the PiPER 7-DoF robotic arm, equipped with a parallel-jaw gripper. This hardware platform closely mimics the kinematic structures and joint compliance seen in common industrial and assistive collaborative robots (cobots).

Physical Fabrication Constraints. To ensure strict adherence to the Non-Printability Score (NPS) constraints, all adversarial patches were exported as high-resolution (300 DPI) PDFs and printed using a standard commercial HP Tank 672 printer utilizing standard CMYK ink profiles. To simulate everyday, real-world deployment scenarios, we strictly printed the patches on ordinary A4 office paper. Glossy photo paper was deliberately avoided to minimize unpredictable specular reflections (glare) under direct laboratory lighting, which introduces severe out-of-distribution pixel clipping that is entirely unmodeled by standard continuous noise assumptions.

A.4 Detailed Optimization Hyperparameters

Initialization and Optimizer. The patch parameters δ were initialized with neutral gray values (RGB value 0.5 across all channels) to prevent early clipping and maximize the bidirectional dynamic range for pixel updates. We optimized the patch for $K = 2,500$ iterations using the standard Adam optimizer with a base learning rate of $\alpha = 0.01$. The learning rate was decayed using a Cosine Annealing schedule down to 1×10^{-4} to explicitly prevent gradient overshoot and ensure stable convergence within the sculpted flat local minima.

Macroscopic EoT Distribution (Γ). For each iteration, we sampled a mini-batch of $\mathcal{B} = 16$ clean frames from the offline datasets. The continuous physical transformation distribution encompassed: (1) Scale adjustments uniformly sampled from $[0.75, 1.25]$ to simulate the arm moving closer to or further from the patch; (2) In-plane rotations sampled from $[-25°, 25°]$ to account for base arm orientations; (3) Perspective warping, shifting the four corners of the patch independently by up to 15% of its spatial dimensions to simulate off-axis 3D viewing angles (yaw/pitch); (4) Brightness and contrast multipliers uniformly sampled from $[0.85, 1.15]$ to emulate ambient shadow casting.

Fabricability Constraint Weights. The Total Variation (TV) penalty weight was empirically set to $\lambda_1 = 1.5 \times 10^{-4}$. Excessive TV weight leads to overly simplistic, monochrome patches that lack semantic features, while insufficient weight results in high-frequency static that fails to survive real-world motion blur. The Non-Printability Score (NPS) weight was set to $\lambda_2 = 2.0 \times 10^{-3}$. The color gamut set C for NPS was populated with 10,000 discrete, physically reproducible RGB tuples densely sampled from the standard Fogra39 CMYK ICC profile.

B Extended Related Work and Differential Analysis

Adversarial Attacks in Robotics. With the widespread application of Vision-Language-Action (VLA) models in end-to-end robotic control, their adversarial vulnerabilities have increasingly become a research focus. Because VLA models directly map cross-modal inputs to low-level physical actions, conventional digital-domain attacks often fail as they disregard the robot's kinematic constraints. To address this, recent studies [Wang et al., 2025a, Lu et al., 2026] have designed physical attack objectives tailored to the geometric characteristics and degrees of freedom (DoFs) of robotic arms. By maximizing spatial and action discrepancies (e.g., UADA, UPA) or directionally manipulating the end-effector trajectory (TMA), these methods successfully expose the execution vulnerabilities of VLAs in the physical space.

Inverting Sharpness-Aware Minimization (SAM). Originally introduced by Foret et al. [2021], Sharpness-Aware Minimization (SAM) is a regularization technique designed to improve the generalization of deep neural networks during standard training. By explicitly penalizing sharp minima in the weight space, SAM ensures that model weights converge to flat regions, thereby reducing overfitting. A few recent works have attempted to apply SAM to adversarial training to build more robust classifiers. Our framework fundamentally inverts and repurposes SAM for offensive physical deployment. First, standard SAM perturbs the model weights θ. An attacker targeting a deployed physical robot has no white-box access to modify the weights of a frozen VLA. P-SAM theoretically reformulates the objective to operate strictly within the input perturbation space δ. Second, existing digital flat-minima techniques apply perturbations homogeneously. In contrast, P-SAM computes a conditionally worst-case approximation $\epsilon^*(x, \tau)$ that is dynamically tied to the macroscopic spatial transformation τ at that exact iteration. This means P-SAM acts as an ensemble flat-minima seeker; it actively sculpts a loss landscape that is universally flat across all possible physical viewpoints the moving robot might traverse, rather than just locally flat in a singular static frame.

C Theoretical Insights into Vulnerability Basin Sculpting

To further substantiate the empirical success of P-SAM, we provide a deeper theoretical intuition regarding the mechanics of adversarial flickering, the justification for our Taylor approximation, and the critical necessity of opaque masking.

C.1 The Mathematical Mechanics of Adversarial Flickering

In standard adversarial patch generation, optimizers seek to minimize the objective $\mathcal{L}_{base}(\delta)$ with respect to the patch pixels δ. Mathematically, standard Empirical Risk Minimization (ERM) inevitably converges to a local minimum $\hat{\delta}$ characterized by a sharp valley. At this minimum, the Hessian matrix $H = \nabla_{\delta}^2 \mathcal{L}_{base}(\hat{\delta})$ exhibits extremely large positive eigenvalues. In a physical robotic deployment, the true observation is modulated by physical sensor noise and microscopic motion blur, denoted as a noise vector η. The effective patch fed to the VLA becomes $\hat{\delta} + \eta$. According to the second-order Taylor expansion:

$$\mathcal{L}_{base}(\hat{\delta} + \eta) \approx \mathcal{L}_{base}(\hat{\delta}) + \eta^\top \nabla_\delta \mathcal{L}_{base}(\hat{\delta}) + \frac{1}{2}\eta^\top H \eta. \tag{9}$$

Since the first-order derivative $\nabla_\delta \mathcal{L}_{base}(\hat{\delta}) \approx 0$ at the local minimum, the loss shift is dominated by the second-order term. Because the eigenvalues of H are massive at a sharp minimum, even an imperceptible microscopic physical noise vector η causes the adversarial loss to surge quadratically, instantly neutralizing the attack and allowing the robot to self-correct. This mathematically formalizes the phenomenon of adversarial flickering.

C.2 Justification of the First-Order Taylor Approximation in P-SAM

In our minimax formulation (Equation 3), finding the exact worst-case microscopic noise ϵ that maximizes the base loss within an ℓ_2 ball of radius ρ is a highly non-convex inner optimization problem. Solving this via iterative Projected Gradient Descent (PGD) at every single step of the outer optimization would multiply the computational cost by a factor of N, rendering patch generation prohibitively slow for 7B+ parameter models. Assuming the loss landscape is locally smooth within the highly constrained microscopic radius ρ, we approximate the perturbed loss using a first-order Taylor expansion:

$$\mathcal{L}_{base}(\delta + \epsilon, x, \tau) \approx \mathcal{L}_{base}(\delta, x, \tau) + \epsilon^\top \nabla_\delta \mathcal{L}_{base}(\delta, x, \tau). \tag{10}$$

To maximize this linear approximation subject to $\|\epsilon\|_2 \leq \rho$, the optimal solution ϵ^* strictly aligns with the direction of the local gradient. This elegantly yields our closed-form approximation in Equation 4. Because ρ represents subtle physical hardware noise rather than global structural semantic shifts, the local linearity assumption holds remarkably well, allowing P-SAM to dynamically track the worst-case degradation with minimal computational overhead.

C.3 Physical Realizability Constraints of the Dynamic Opaque Mask

A critical innovation in our Macroscopic Kinematic Simulation (Section 2.1) is the mathematical enforcement of the dynamic opaque mask $M_\tau \in \{0, 1\}^{H \times W}$. Rather than acting merely as a spatial bounding box, M_τ serves as the mathematical guarantor of physical realizability. To distinguish a legitimate localized physical attack from an impossible global digital perturbation, M_τ enforces two strict physical constraints:

(1) Strict Opaque Occlusion: Standard digital attacks typically utilize an additive perturbation formulation $x_{adv} = x + \delta$ or alpha-blending $x_{adv} = (1 - \alpha)x + \alpha\delta$. In the digital domain, this mathematically models a translucent, "ghost-like" overlay, allowing the optimizer to inadvertently exploit residual background pixels (e.g., the texture of a wooden table or the edges of the target object) underlying the patch to minimize the loss. However, in the physical world, a printed piece of paper is 100% opaque ($\alpha = 1$). This geometric discrepancy destroys the adversarial gradient alignment upon physical printing, causing immediate sim-to-real failure. To resolve this, our objective $o(x, \delta, \tau) = (1 - M_\tau) \odot x + M_\tau \odot \mathcal{T}(\delta, \tau)$ acts as a precise spatial carving mechanism. The term $(1 - M_\tau) \odot x$ completely zeroes out the background pixels before patch insertion. This forces the optimizer to achieve targeted task failure utilizing solely the information capacity of the patch pixels themselves, ensuring theoretical fidelity between the simulated optimization state and the physical deployment state.

(2) Dynamic Geometric Synchronization: As the robot's camera translates through 3D space, the 2D projection of the physical patch continuously distorts (e.g., a square sticker viewed from an acute angle projects as a trapezoid or rhombus). By subjecting the base binary mask M to the exact same macroscopic transformation τ, the resulting mask M_τ dynamically tracks the valid geometric boundary of the perturbation. This strict synchronization prevents the introduction of out-of-bound digital artifacts (such as the zero-padding artifacts inherently introduced by differentiable spatial transformations during affine rotations or perspective warps), ensuring the adversarial gradients are strictly confined to the physical dimensions of the deployable sticker.

D Supplementary Experiments

D.1 Resilience Against Algorithmic Input Defenses

A standard, intuitive countermeasure against visual adversarial attacks involves input-level preprocessing. As shown in Figure 9, to evaluate the robustness of our framework against such mitiga-

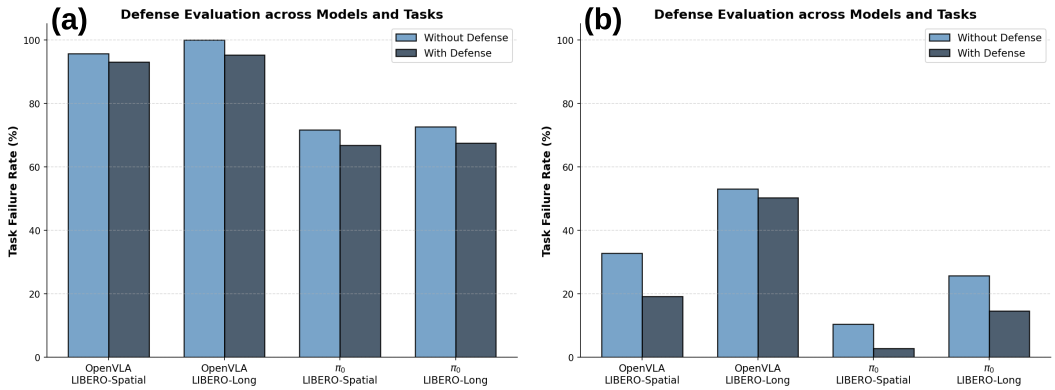

Figure 9: Task Failure Rates (TFR) under low-pass algorithmic defenses. (a) Our P-SAM maintains high efficacy. (b) Conventional unconstrained baseline suffers a precipitous drop.

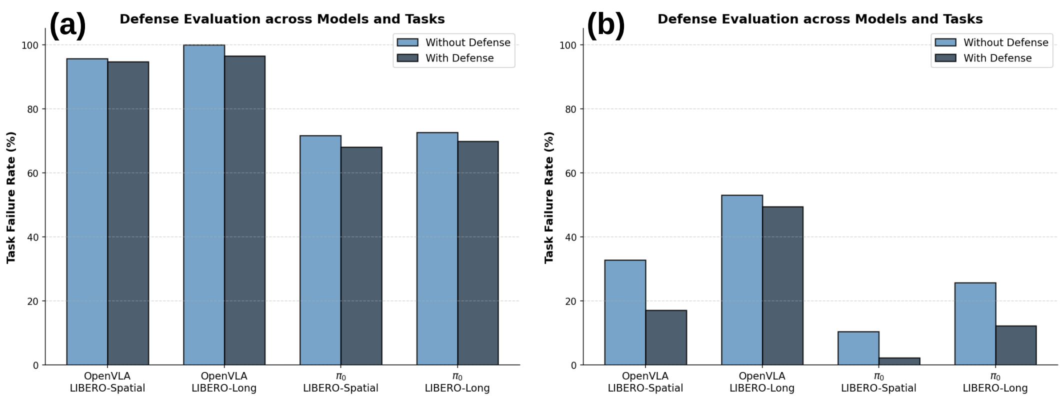

Figure 10: Task Failure Rates (TFR) under explicit semantic defense instructions. (a) Our P-SAM efficiently overrides semantic guardrails. (b) Conventional unconstrained baseline.

tions, we subjected the VLA's visual observations to conventional image transformations in simulation, specifically JPEG compression and Gaussian blurring. Because conventional unconstrained baselines heavily rely on brittle, high-frequency pixel artifacts, these simple low-pass algorithmic defenses easily neutralized the adversarial signals, triggering a precipitous drop in their Task Failure Rate (TFR). In contrast, the P-SAM patch demonstrated inherent resilience to these visual interventions, benefiting from organic and locally uniform macroscopic structures forged jointly by the $\mathcal{L}_{TV}$ constraint and minimax optimization. This empirical evidence confirms that actively sculpting a flat-minima geometry fundamentally circumvents standard filtering-based robustification techniques.

D.2 Visual Hijacking vs. Explicit Semantic Instructions

Given that VLAs inherently rely on multimodal vision-language alignment, we investigated whether explicit semantic priors could act as a cognitive defense to override our physical visual attack. As shown in Figure 10, we designed a semantic override scenario where the VLA was commanded via natural language with explicit textual guardrails: "Pick up the target object, and strictly ignore the colored patch on the table." Strikingly, the P-SAM patch successfully induced a state of "language deafness." The VLA completely disregarded the explicit semantic defense and compulsively executed the maliciously injected kinematic trajectory. This exposes a profound modality dominance vulnerability in current embodied foundation models: persistent, maliciously optimized visual stimuli can overwhelmingly dominate and bypass language-conditioned safety alignments, dictating physical actions regardless of semantic constraints.

E Real-World Evaluation Scenarios

Object Grasping (OBJ): As the most fundamental primitive (e.g., “Pick up the blue cube”), this task requires precise depth estimation and target localization. The adversarial patch is placed adjacent to the target object. The goal is to corrupt the VLA’s spatial reasoning. Successful attacks typically manifest as the VLA grasping thin air, closing the gripper prematurely, or forcefully colliding with the table.

Object Placement (PLC): This task involves both picking and complex spatial positioning (e.g., “Place the block into the bowl”), demanding robust hand-eye coordination. The patch continuously perturbs the spatial understanding of the receptacle, frequently forcing the robot to release the object entirely outside the bowl or deliberately drop it off the edge of the workspace.

Complex Manipulation (MNP): This task involves articulated objects and continuous environmental contact (e.g., “Push the sliding door to the left”). Adversarial flickering is most prominent here, as the physical contact dynamics and mechanical friction cause severe micro-vibrations in the camera mount. Standard static patches fail completely under these vibrations, whereas P-SAM’s flat basin seamlessly absorbs this hardware jitter, maintaining a steady adversarial drift that causes the robot to miss the handle entirely.

F Broader Impacts and Potential Defensive Paradigms

F.1 Dual-Use Nature and Ethical Disclosure

The development of physical-world adversarial attacks against robotic systems inherently carries dual-use implications. As VLA foundation models accelerate the deployment of autonomous agents in unstructured human environments (e.g., domestic service robots, automated manufacturing, and healthcare assistants), ensuring their operational safety becomes a life-critical imperative.

This research explicitly highlights a profound, previously underappreciated vulnerability: closed-loop robotic systems can be completely hijacked, overriding safety protocols and semantic instructions, simply by passively placing a carefully generated, printable sticker in their visual workspace. While our P-SAM framework provides a highly effective blueprint for generating physical attacks without requiring digital access to the robot’s network, this serves as a necessary “red-teaming” effort. We disclose these vulnerabilities strictly for academic auditing, robustness evaluation, and defensive research purposes, aiming to catalyze the development of secure robotic systems before widespread commercial integration occurs.

F.2 Proposed Defensive Countermeasures

Adversarial Pre-Training in Action Spaces. Current foundation models are trained exclusively on pristine trajectory datasets (e.g., Open X-Embodiment). Incorporating P-SAM-generated patches directly into the massive pre-training loop as a data augmentation technique could force the VLA’s vision encoder to learn to semantically disentangle high-frequency adversarial noise from valid task-relevant geometric features. However, scaling this to billions of parameters currently presents a severe computational bottleneck.

Multimodal Sensory Fusion and Cross-Validation. Over-reliance on single-modality RGB sensors (a single wrist or third-person camera) creates a critical single point of failure. Integrating a multi-camera fusion setup could allow the robotic system to cross-verify the 3D geometry of the scene. An adversarial patch is highly unlikely to perfectly satisfy the worst-case flat-minima optimization for two distinctly separated physical viewpoints simultaneously. Furthermore, fusing visual inputs with active depth sensing (LiDAR) or tactile feedback would drastically increase the difficulty for an attacker to generate a viable physical override.

G More Visual Presentations

More attack examples. Figure 11 shows more successful results of our attacks, while Figure 12 presents a failed result. The model we have deployed here is OpenVLA [Kim et al., 2024]. As illustrated in Section 5, severe topological occlusions of the camera’s line-of-sight emerge as a highly

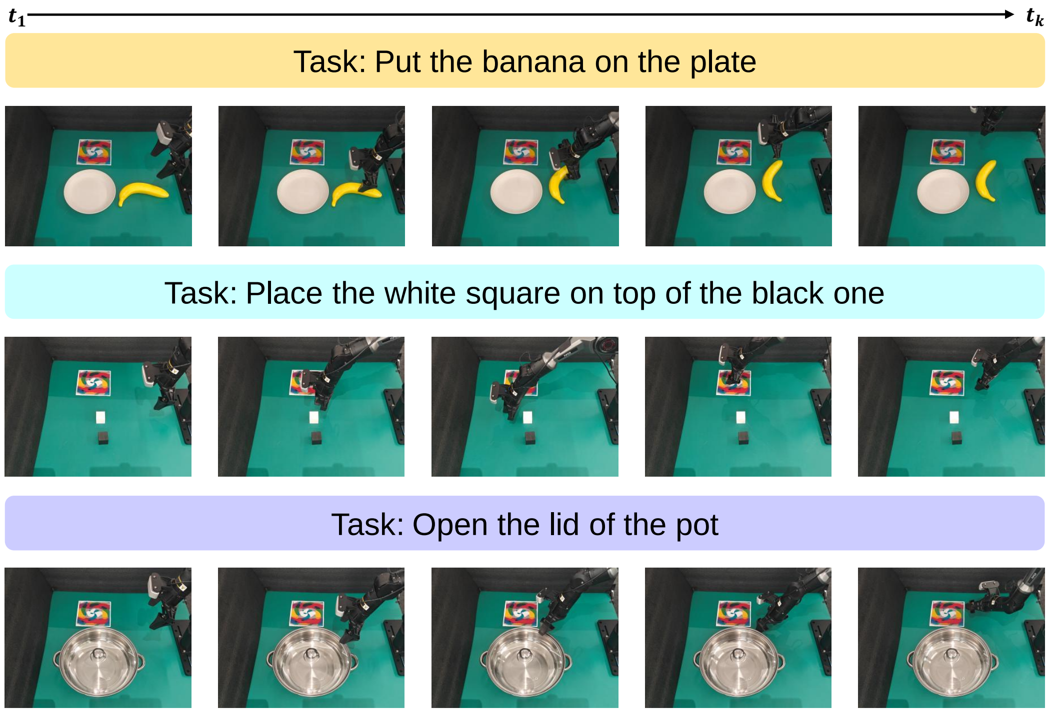

Figure 11: Examples of successful physical attacks across different manipulation tasks.

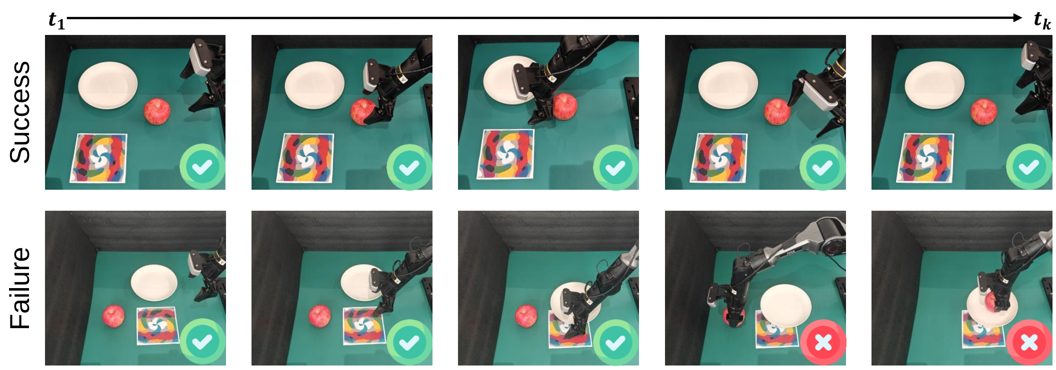

Figure 12: An example of a failed attack rollout. The task objective is to put the apple on the plate. We can observe that when the robotic arm accidentally covers the adversarial patch with the plate, the transmission of adversarial visual features is severed, causing the attack to fail.

probable cause of physical attack failures. Because closed-loop systems rely on continuous visual streaming to update action predictions, the adversarial influence of the P-SAM patch demands an uninterrupted visual pathway. Momentary loss of visibility abruptly halts the injection of worst-case pixel perturbations into the vision encoder, rendering the sculpted flat vulnerability basins inaccessible. Consequently, the adversarial manipulation sequence is suspended, allowing the model to momentarily recover its benign control policy and break the hijacking loop. This phenomenon confirms that while P-SAM successfully mitigates algorithmic adversarial flickering, continuous geometric visibility remains a strict, physical prerequisite for reliably executing the attack in the real world.

More patch examples. Figure 13 visualizes a selection of our generated adversarial patches. As can be observed, our patches exhibit distinct, organic macroscopic structures. This unique visual appearance is a direct consequence of our optimization framework: the Total Variation ($\mathcal{L}_{TV}$) constraint, coupled with the microscopic minimax optimization (P-SAM), jointly forge these piecewise-

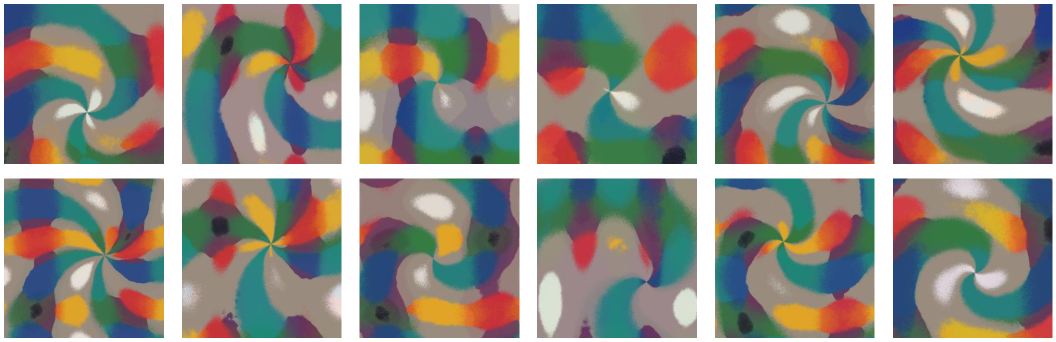

Figure 13: Examples of generated P-SAM adversarial patches. The target model is OpenVLA.

constant macro-patterns. By sculpting these flat vulnerability basins, our patches inherently resist common image-space interventions in the physical world (such as lighting variations or viewpoint deformations), ensuring robust deployability and high transferability across different VLA architectures.